

GEOG 28IA

Space and Place

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Learning Objectives: To understand alternative conceptualization of space and place used, or not used, in GIScience. To begin to understand how those conceptualization interact with the computational systems beneath common GISystems.



GOOD OMENS

O'Sullivan (ch. 8) frames simulation through Alfred North Whitehead's view of reality as an ongoing process and engages with the idea of flat ontology, both of which challenge static, reductionist representations of the world. However, these metaphysical frameworks offer little guidance for implementation, leaving researchers to grapple with how they can be used in simulation models. Because fully representing a process-based reality would require accounting for a near infinite set of variables, a "complete" simulation remains unattainable. To what extent should researchers strive to approximate Whitehead's definition of reality in their models? Is it sufficient to acknowledge the inherent limitations of simulations, or is there an ethical responsibility to continually refine models toward greater ontological accuracy, even if perfect representation is impossible?

O'Sullivan (Ch. 8) argues there is no one-to-one mapping between patterns and processes, and identifies simulation as the computational tool most aligned with process thinking. Yet in the same chapter he concedes that many things are permanent enough that it is "pragmatically useful" to treat them as things. This feels like a tension: on one hand, pattern-based analysis can't reliably tell us what process produced a pattern; on the other, treating the world as made of stable entities is just too convenient to give up. Is this a reasonable pragmatic compromise, or does it give GIScience a permanent excuse to stay comfortable with form and never seriously take on the process-oriented challenges that O'Sullivan himself raises?

How much of the emergent spatial patterns seen at a "space" scale can be explained by the relational processes embedded in particular "places"?

ATOMS AND SPACES

PROMPT

Outline the *geoatom* as presented in O'Sullivan and originally discussed by Goodchild (1990) and Goodchild et al. (2007)

PROMPT

What are difference conceptualization of space identified within the geographic literature?

- Define a conceptualization of space.
- Present examples of that conceptualization in GIScience applications or techniques

ACTIVITY

Purpose: To link different types of space to a major research initiative working to solve an important applied problem and advance GIScience.

Task: Create a list of way different types of space may be important when examining the research question driving the Wildfire Resilience Initiative.

BREAK

DEFINING PLACE

PROMPT

Work together to trace the ideas that contribute geographers and GIScientists conceptualizations and definitions of place.

BREAK

COMPUTABLE SPACE

PROMPT

How do GISystems accommodate different conceptualizations of space and place? How are those concepts actually fit into computer systems?

OPEN QUESTION

How do we avoid moving from the rigidity of formalism to the opposite problem of arbitrariness and impressionistic interpretation? In other words, when these materials criticize existing GIS representations, are they really pointing toward a more rigorous scientific program, or mainly toward a broader philosophical unease with formal representation itself?